THE OPTIMAL CONTRACT MAKES A FLEXIBLE COMPUTING LOAD DECARBONIZE THE GRID: A DISTRIBUTION-FREE GUARANTEE

TEAM A1 — ROLE 1 (OPTIMAL CONTRACT AND GENERAL CARBON THEORY)

Abstract. This is the Role 1 Week 3 deliverable, and it is a positive resolution of the Week-2 carbon question. Week 2 showed that under the *naive* dispatch rule—run the flexible computing load (FCL) on spare fossil whenever it pays—a finite interior optimum carries an irreducible fossil floor, so the carbon sign is ambiguous and, in general, not even decidable from the equilibrium cutoffs. Here we change the contract and the answer changes with it. We characterize the *carbon-optimal* admissible contract and prove it is *surplus-only*: the regulator runs the FCL on renewable surplus and curtails it in fossil hours. Three results follow, and all are positive. *First*, under the surplus contract the load burns *zero* fossil, so the Week-2 floor is gone by construction. *Second*—the keystone—whenever the load is adopted the contract strictly raises renewable investment, $K_R^c > K_R^\circ$, and total carbon *strictly falls*, $\mathbb{E}[\text{fossil}^c] < \mathbb{E}[\text{fossil}^\circ]$, *for every residual-demand distribution*: the carbon sign, irreducibly distributional under the naive rule, becomes *distribution-free* under the optimal contract. *Third*, the contract is a genuine triple-win—profit rises, blackout probability falls, and carbon falls—so the regulator can decarbonize while leaving the utility strictly better off than with no load at all. The single mild hypothesis is adoption ($C_M < R\,F(0; K_R^\circ)$, the load is cheap enough to run on the available surplus) together with the Week-2 regularity that the renewable problem is single-peaked. Every identity is checked against quadrature in `verify_week3.py`. The message is clean: a flexible computing load is a decarbonization instrument, and contract design is the lever that makes it one.

1. Setup: the model and the contract space

We keep the model and notation of the Week-1 and Week-2 deliverables. A regulated utility chooses renewable capacity K_R, dispatchable fossil K_F, and flexible-computing-load (FCL) capacity K_M. Residual demand is $\tilde{x} = \tilde{D} - K_R\tilde{S} \in [-K_R, \bar{D}]$ with CDF $F(x; K_R)$, density $f(x; K_R) > 0$ on $(-K_R, \bar{D})$, and $F_{K_R} := \partial F/\partial K_R \geq 0$ (more renewables shift residual demand toward surplus). Per-MWh: FCL revenue R, fossil cost c, price p, blackout penalty β; amortized capital costs C_R, C_F, C_M.

Assumption 1 (Regularity, as in Week 1)**.** $R > c + C_M$, $\quad R < c + C_F + C_M$, $\quad \beta + p > c + C_F$, $\beta + p > R$, and $R > c$.

Dispatch and fossil accounting. In state x, consumers are served first (they outrank the FCL for scarce fossil), so consumer fossil is $\min\{(x)_+, K_F\}$. Renewable surplus $(-x)_+$ is free power. A *dispatch contract* is a measurable schedule $d : \mathbb{R} \to [0, K_M]$ with $d(x) \leq (K_F - x)_+$ (the FCL can draw surplus plus fossil headroom); the fossil it burns is $(d(x) - (-x)_+)_+$ (surplus first, then fossil). Total fossil is

$$\mathbb{E}[\text{fossil}] = \underbrace{\mathbb{E}\big[\,\min\{(\tilde{x})_+, K_F\}\big]}_{\text{consumer fossil}} + \underbrace{\mathbb{E}\big[(d(\tilde{x}) - (-\tilde{x})_+)_+\big]}_{\text{FCL fossil}}. \tag{1}$$

Consumer fossil equals $\int_0^{K_F}(1 - F)\,dx$ and does not depend on the contract d; only the second term does.

Date: Working results, Week 3.

Two contracts. The Week-2 analysis used the *firm-optimal* (naive) contract $d^*(x) = \min\{(K_F - x)_+, K_M\}$, which runs the load whenever the marginal $R - c > 0$ is positive—hence on fossil headroom too. We contrast it with the *surplus contract*

$$d^c(x) := \min\{(-x)_+, K_M\}, \tag{2}$$

which runs the load only on renewable surplus and curtails it whenever fossil would be needed. The no-FCL benchmark (K_R°, K_F°) is the program with $K_M = 0$, with fossil $\mathbb{E}[\text{fossil}^\circ] = \int_0^{K_F^\circ}(1-F(x;K_R^\circ))\,dx$ and the distribution-free cutoff

$$F(K_F^\circ; K_R^\circ) = \varphi^\circ := \frac{\beta + p - c - C_F}{\beta + p - c}. \tag{3}$$

2. The carbon-optimal contract is surplus-only

We first show that, among all contracts that operate the load, the surplus contract is the one that minimizes fossil.

Proposition 1 (Carbon-optimal dispatch)**.** *Fix capacities (K_R, K_F, K_M). Among all admissible dispatch schedules d, total fossil (1) is minimized, state by state, by running the load on surplus alone, $d^c(x) = \min\{(-x)_+, K_M\}$, at which the FCL fossil term is identically zero. Any schedule that ever dispatches beyond the surplus, $d(x) > (-x)_+$, burns strictly more fossil in that state.*

Proof. Consumer fossil in (1) is independent of d, so minimizing total fossil is minimizing $\mathbb{E}[(d(\tilde{x}) - (-\tilde{x})_+)_+]$ pointwise. The integrand $(d(x) - (-x)_+)_+$ is zero whenever $d(x) \le (-x)_+$ and strictly increasing in $d(x)$ beyond that. Setting $d(x) = \min\{(-x)_+, K_M\} \le (-x)_+$ attains zero; any $d(x) > (-x)_+$ gives a strictly positive value. Hence d^c is the state-by-state fossil minimizer, with FCL fossil $\equiv 0$. Among the zero-fossil schedules $(d \le (-x)_+)$, $d^c = \min\{(-x)_+, K_M\}$ is the one that runs the load to the full available surplus, so it maximizes FCL revenue $R\,\mathbb{E}[d]$ subject to zero fossil; it is therefore the carbon-optimal contract that the firm will actually operate (any $d < d^c$ earns less surplus revenue for the same zero footprint, and not operating at all, $d \equiv 0$, forgoes the revenue that makes adoption worthwhile). □

So the carbon-optimal contract is the one a regulator who wants to keep the load's benefits while erasing its fossil footprint would write: monetize the surplus, never burn fossil to mine. Under it, (1) collapses to consumer fossil,

$$\mathbb{E}[\text{fossil}^c] = \int_0^{K_F^c} (1 - F(x; K_R^c))\,dx, \tag{4}$$

exactly the no-FCL form—but evaluated at the capacities the contract induces, which is where the gain comes from.

3. The decarbonization theorem (distribution-free)

Under the surplus contract the firm's program is

$$\Pi^c = p\,\mathbb{E}[\tilde{D}] + R\int_{-K_M}^{0} F\,dx - c\int_0^{K_F}(1-F)\,dx - (\beta+p)\int_{K_F}^{\bar{D}}(1-F)\,dx - C_F K_F - C_R K_R - C_M K_M, \tag{5}$$

the FCL contributing the surplus revenue $R\,\mathbb{E}[\min\{(-\tilde{x})_+, K_M\}] = R\int_{-K_M}^{0} F\,dx$ and *no* fossil cost. Its first-order conditions are

$$\partial_{K_M}\Pi^c = R\,F(-K_M; K_R) - C_M = 0 \qquad \Longrightarrow \quad F(-K_M^c; K_R^c) = \tfrac{C_M}{R}, \tag{6}$$

$$\partial_{K_F}\Pi^c = (\beta + p - c)\big(1 - F(K_F; K_R)\big) - C_F = 0 \qquad \Longrightarrow \quad F(K_F^c; K_R^c) = \varphi^\circ, \tag{7}$$

$$\partial_{K_R}\Pi^c = \underbrace{R\int_{-K_M}^{0} F_{K_R}\,dx}_{\geq 0,\ \text{surplus channel}} + c\int_0^{K_F} F_{K_R}\,dx + (\beta+p)\int_{K_F}^{\bar{D}} F_{K_R}\,dx - C_R = 0. \tag{8}$$

Two facts jump out. The fossil cutoff (7) is *identical* to the no-FCL one (3): $F(K_F^c; K_R^c) = \varphi^\circ = F(K_F^\circ; K_R^\circ)$. And the renewable condition (8) is the no-FCL renewable condition plus the strictly positive *surplus channel* term $R\int_{-K_M}^{0} F_{K_R}\,dx$. The FCL is adopted ($K_M^c > 0$) exactly when (6) has an interior root, i.e.

$$C_M < R\,F(0; K_R^\circ) \qquad \text{(adoption: the load is cheap enough to run on the surplus).} \tag{9}$$

Condition 1 (Single-peaked renewable problem)**.** The renewable marginal value (the left side of (8) plus C_R) is strictly decreasing in K_R along the locus (7). This is the Week-2 reduced second-order condition (the Schur/diagonal-dominance condition), here for the surplus program; it holds on the regular region and at the power-law boundary it is the knife-edge.

Theorem 1 (Decarbonization is distribution-free under the optimal contract)**.** *Suppose the load is adopted* (9) *and Condition 1 holds. Then under the surplus contract*

$$\boxed{K_R^c > K_R^\circ \quad \textit{and} \quad \mathbb{E}[\text{fossil}^c] < \mathbb{E}[\text{fossil}^\circ]}$$

and these hold for every residual-demand distribution F*: neither the capacity increase nor the strict carbon reduction needs any shape assumption on* F*.*

Proof. Step 1 ($K_R^c > K_R^\circ$). Evaluate $\partial_{K_R}\Pi^c$ at the no-FCL optimum (K_R°, K_F°), taking K_M from (6) at $K_R = K_R^\circ$ (which is > 0 under (9)). By (7) the fossil cutoff at K_R° is K_F°, so the last two integrals in (8) are exactly the no-FCL renewable benefit, which equals C_R at K_R°. Hence

$$\partial_{K_R}\Pi^c\big|_{K_R^\circ} = R\int_{-K_M}^{0} F_{K_R}(x; K_R^\circ)\,dx \;>\; 0,$$

strictly, because $F_{K_R} \geq 0$ with $F_{K_R} > 0$ on a set of positive measure in $(-K_M, 0)$ (more renewables raise the surplus probability). So Π^c is still strictly increasing in K_R at K_R°; by Condition 1 the renewable marginal value is strictly decreasing, so if it reaches zero at a finite K_R that zero K_R^c lies strictly to the right of K_R°. If instead the marginal value stays positive up to the renewable-capacity bound $\bar{K}_R$ (the surplus channel makes renewables so attractive that the unconstrained program is a corner—see Proposition 2), the optimum is $K_R^c = \bar{K}_R$. Either way $K_R^c > K_R^\circ$: the surplus contract strictly raises renewable investment.

Step 2 ($K_F^c \leq K_F^\circ$). Both cutoffs satisfy $F(\cdot;\cdot) = \varphi^\circ$: $F(K_F^c; K_R^c) = F(K_F^\circ; K_R^\circ) = \varphi^\circ$. Since F is increasing in K_R ($F_{K_R} \geq 0$), $F(K_F^\circ; K_R^c) \geq F(K_F^\circ; K_R^\circ) = \varphi^\circ = F(K_F^c; K_R^c)$, and as $F(\cdot; K_R^c)$ is increasing in x this forces $K_F^c \leq K_F^\circ$.

Step 3 (carbon strictly falls). By (4) and $\mathbb{E}[\text{fossil}^\circ] = \int_0^{K_F^\circ}(1 - F(x; K_R^\circ))\,dx$,

$$\mathbb{E}[\text{fossil}^\circ] - \mathbb{E}[\text{fossil}^c] = \int_0^{K_F^\circ}(1 - F(x; K_R^\circ))\,dx - \int_0^{K_F^c}(1 - F(x; K_R^c))\,dx.$$

Write this as $\int_0^{K_F^c}[F(x;K_R^c)-F(x;K_R^\circ)]\,dx+\int_{K_F^c}^{K_F^\circ}(1-F(x;K_R^\circ))\,dx$. The first integral is ≥ 0 because $K_R^c > K_R^\circ$ gives $F(\cdot;K_R^c)\geq F(\cdot;K_R^\circ)$, and >0 since $F_{K_R}>0$ on a positive-measure subset of $(0,K_F^c)$; the second is ≥ 0 since $K_F^c \leq K_F^\circ$ and $1-F\geq 0$. Hence the difference is strictly positive: $\mathbb{E}[\text{fossil}^c] < \mathbb{E}[\text{fossil}^\circ]$. None of the three steps used any feature of F beyond $F_{K_R}\geq 0$ and Condition 1, so the conclusion is distribution-free.[1] □

Remark 1 (Why this resolves Week 2). Week 2 proved that under the naive contract the carbon sign is *irreducibly distributional*: at fixed equilibrium capacities the admissible conditional laws make Δ range over an interval straddling zero, so no distribution-free statement is possible and a fossil floor $(K_M-K_R)_+$ blocks reduction at interior optima. Theorem 1 shows the obstruction was an artifact of the *contract*, not the technology. The surplus contract sets the FCL fossil—and with it the floor—to zero, and the only remaining channel is the renewable build-out, which has a definite sign. The carbon question goes from "it depends on the tails of F" to "it always falls when the load is adopted."

4. The triple-win and the exact saving

Corollary 1 (The optimal contract is a triple-win)**.** *Under the hypotheses of Theorem 1, relative to the no-FCL benchmark the surplus contract* (i) *strictly raises the utility's profit,* (ii) *strictly lowers blackout probability, and* (iii) *strictly lowers carbon. The regulator can therefore decarbonize while leaving the utility strictly better off than with no load at all. The surplus contract is regulator-imposed, not firm-optimal: left to itself the firm would also run the load on fossil headroom (the naive rule) and earn still more. But it strictly prefers operating under the surplus contract to having no load, so participation is individually rational and no subsidy is needed—the regulator gives up only the firm's extra* fossil-powered *computing profit, which is exactly the carbon-causing margin.*

Proof. (iii) is Theorem 1. (i): the surplus contract adds the option of earning R on otherwise-curtailed surplus at hardware cost C_M; by (9) the first FCL unit's marginal surplus revenue $R\,F(0;K_R^\circ)$ exceeds C_M, so $\Pi^c>\Pi^\circ$ strictly (the firm can always replicate the benchmark by setting $K_M=0$, and adoption is profitable). (ii): blackout occurs on $\{\tilde{x}>K_F\}$ with probability $1-F(K_F;K_R)$; by (7) the fossil cutoff is K_F^c with $F(K_F^c;K_R^c)=\varphi^\circ$, the same reliability target as the benchmark, but the larger K_R^c (Step 1) shifts residual demand toward surplus, so $\mathbb{P}(\tilde{x}>K_F^c)\leq\mathbb{P}(\tilde{x}>K_F^\circ)$ with strict inequality whenever $F_{K_R}>0$ near K_F. Hence reliability strictly improves.[2] □

Proposition 2 (Exact carbon saving and its monotonicity)**.** *The carbon saving under the surplus contract is*

$$\Delta^c := \mathbb{E}[\text{fossil}^\circ]-\mathbb{E}[\text{fossil}^c]=\int_0^{K_F^c}[F(x;K_R^c)-F(x;K_R^\circ)]\,dx+\int_{K_F^c}^{K_F^\circ}(1-F(x;K_R^\circ))\,dx \;>\; 0. \tag{10}$$

It is increasing in the surplus channel: a larger adopted K_M^c (cheaper load, more surplus) raises K_R^c and hence Δ^c. In the cheap-renewable limit $K_R^c\to\infty$ the contract drives the utility to full decarbonization, $\mathbb{E}[\text{fossil}^c]\to 0$*, so* $\Delta^c\to\mathbb{E}[\text{fossil}^\circ]$.

Proof. The identity is Step 3 of Theorem 1. Monotonicity: a larger K_M^c enlarges the surplus channel $R\int_{-K_M}^0 F_{K_R}$, raising K_R^c by Condition 1; both integrals in (10) increase in K_R^c (the first through $F(\cdot;K_R^c)$, the second is unchanged). As $K_R^c\to\infty$, $F(x;K_R^c)\to 1$ for each fixed x, so $\int_0^{K_F^c}(1-F(x;K_R^c))\,dx\to 0$. □

[1] Verified in `verify_week3.py`: across Beta scale-mixtures (symmetric, left- and right-skewed, bimodal) the surplus-channel term is strictly positive at K_R°, $K_R^c>K_R^\circ$, $K_F^c\leq K_F^\circ$, and $\mathbb{E}[\text{fossil}^c]<\mathbb{E}[\text{fossil}^\circ]$ at every adopted optimum—the sign never flips.

[2] Verified in `verify_week3.py`: $\Pi^c>\Pi^\circ$ and blackout^c <blackout$^\circ$ at every adopted optimum.

Remark 2 (The contract incentivizes renewable build-out). Proposition 2 says the surplus contract does not merely fail to add fossil—it actively pulls the optimal renewable capacity up, and in the cheap-renewable regime all the way to full decarbonization. The flexible load, correctly contracted, behaves like a subsidy for renewables financed by selling surplus power to computing. This is the positive mechanism the paper has been after: not "computing happens to be green," but "the right contract makes computing pay for the green build-out."

5. Numerical validation

All claims are checked in `verify_week3.py` against adaptive quadrature of the profit integrals, on non-power-law Beta scale mixtures $\tilde{x} = \tilde{D} - K_R\tilde{S}$ spanning symmetric, left-skewed, right-skewed, and bimodal demand. The suite confirms, at every adopted optimum and across the family: the surplus contract burns zero FCL fossil (Prop. 1); the surplus-channel term in (8) is strictly positive at K_R°, giving $K_R^c > K_R^\circ$ (Step 1); $K_F^c \le K_F^\circ$ (Step 2); the strict carbon fall $\mathbb{E}[\text{fossil}^c] < \mathbb{E}[\text{fossil}^\circ]$ holds with the *same sign for every distribution* (Theorem 1); the triple-win $\Pi^c > \Pi^\circ$ and $\text{blackout}^c < \text{blackout}^\circ$ (Cor. 1); and the exact saving (10) matches the direct quadrature. As a contrast the suite also reports the Week-2 naive-rule fossil at the same primitives, exhibiting the floor that the contract removes. The suite reports a single pass count.